Satilmis Furkan KAHRAMAN

Department of Electrical and Electronics Engineering
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RESEARCH INTERESTS

Medical Imaging, Image Processing, Signal Processing, Hardware Design and Implementation

EDUCATION

Bilkent University, Ankara, Turkey
Master of Science, Electrical and Electronics Engineering (English)
August 2025 - June 2028 (Expected) **CGPA: 3.42/4.00**

Bilkent University, Ankara, Turkey
Bachelor of Science, Electrical and Electronics Engineering (English)
August 2020 - June 2025 **CGPA: 3.38/4.00**

PUBLICATION

- S. Furkan Kahraman, Ege Kor, Emine Ulku Saritas, "Single-Sided MPI without Selection Fields: Proof-of-Concept Results," Published in International Journal on Magnetic Particle Imaging (IJMPI), 2025.

HONOR & AWARDS

- Ranked **61st among 2.5 million examinees** in the nationwide university entrance examinations, Turkey, 2020.
- Comprehensive scholarship covering full tuition and monthly stipend for undergraduate education in the top engineering school in Turkey, Bilkent University, Ankara, Turkey, 2020-2025.
- **Research Excellence Award** by Bilkent Electrical and Electronics Engineering Department, 2025.

ACADEMIC EXPERIENCE

Undergraduate Researcher, Aug. 2023 - Present
National Magnetic Resonance Research Center (UMRAM), **Bilkent University**
Advisor: Assoc. Prof. Emine Ulku Saritas
Engaged in research focused on advancing magnetic particle imaging (MPI) techniques for in vivo imaging using magnetic nanoparticles (MNPs). Designed and developed a magnetic particle spectrometer (MPS) device for characterizing the magnetization response of MNPs.

Undergraduate Researcher, Aug. 2022 - Feb. 2023
Ultra Fast Laser Laboratory (UFOLAB), **Bilkent University**
Advisor: Prof. Fatih Omer Ilday
Conducted research on nonlinear feedback-driven laser material interactions for laser sintering techniques, utilizing lasers as power and heat sources to sinter powdered material based on 3D model-defined points, ultimately forming solid structures.

TEACHING EXPERIENCE

Teaching Assistant, Fall 2025 - Spring 2026
Bilkent University
Course: Analog Electronics
Assisted in conducting lab sessions, grading assignments, and holding office hours for undergraduate engineering students.

Teaching Assistant, Fall 2025 - Spring 2026
Bilkent University
Course: Electronic Circuit Design
Supported course instruction, helped evaluate project designs, and provided mentorship on practical circuit implementation.

INTERNSHIPS

Intern, Jun. 2024 - Sep. 2024
Institute of Biomedical Imaging (IBI), **Technical University of Hamburg**
Worked on simulation and design of Magnetic Particle Imaging subsystems. Conducted research to design and implement a 3D Helmholtz/Maxwell coil for magnetic sensor calibration.

Intern, Aug. 2023 - Sep. 2023
National Magnetic Resonance Research Center (UMRAM), **Bilkent University**
Conducted magnetic field simulations to design a magnetic particle spectrometer (MPS) setup. Performed 3D design of the Magnetic Particle Spectrometer setup.

Intern, Jun. 2023 - Jul. 2023
Helicopter Maintenance Factory, **Ministry of National Defense**
Worked on advancing a laser-based distance measurement system, involving a thorough exploration of theoretical advancements, including the development of innovative methodologies and algorithms to enhance precision and accuracy in distance measurements utilizing laser technology.

NOTABLE PROJECTS

MFH-MPS Hybrid System
Graduate Research (M.Sc.)
Combined magnetic fluid hyperthermia (MFH) and magnetic particle spectrometer (MPS) experimental platform, synchronizing AWG-driven heating with real-time temperature and MPS signal acquisition.

Magnetic Nanoparticle Imaging with a Single-Sided MPI
EEE 492 - Individual Research
Depth imaging using a single-sided Magnetic Particle Imaging to detect skin beneath nano-gels without selection fields.

3-Axis Helmholtz Coil
EEE 399 - Summer Internship
Design and implementation of 3D Helmholtz coil for Magnetic Particle Imaging head scanner sensor calibration.

Single-sided Magnetic Particle Spectrometer
EEE 491 - Individual Research
The development and application of a single-sided Magnetic Particle Spectrometer (MPS) device for detecting the response of magnetic nanoparticles (MNPs), for the ultimate goal of non-invasive tracking of the degradation of implanted MNPs/nanogels.

TRC-11 Transreceiver
EEE 211 - Analog Electronics
Design and implementation of a low power consumption receiver-transmitter radio circuit operating at 27 MHz frequency.

Autonomous Aircraft
EEE 102 - Digital Circuit Design
Development and implementation of a drone capable of maintaining balance in the air using FPGA, with automatic takeoff and landing features.

COMPUTER SKILLS

Languages: MATLAB, Python, C++, Julia, VHDL, Assembly for Intel 8051, Spice Tools, Office Tools
Applications: VSCode, COMSOL, Autodesk, Lightroom

LANGUAGES

Languages: English (IELTS Academic: 7), German (Beginner), Turkish (Native)